

0.17

Fraud Rate %

506

CRITICAL Flagged

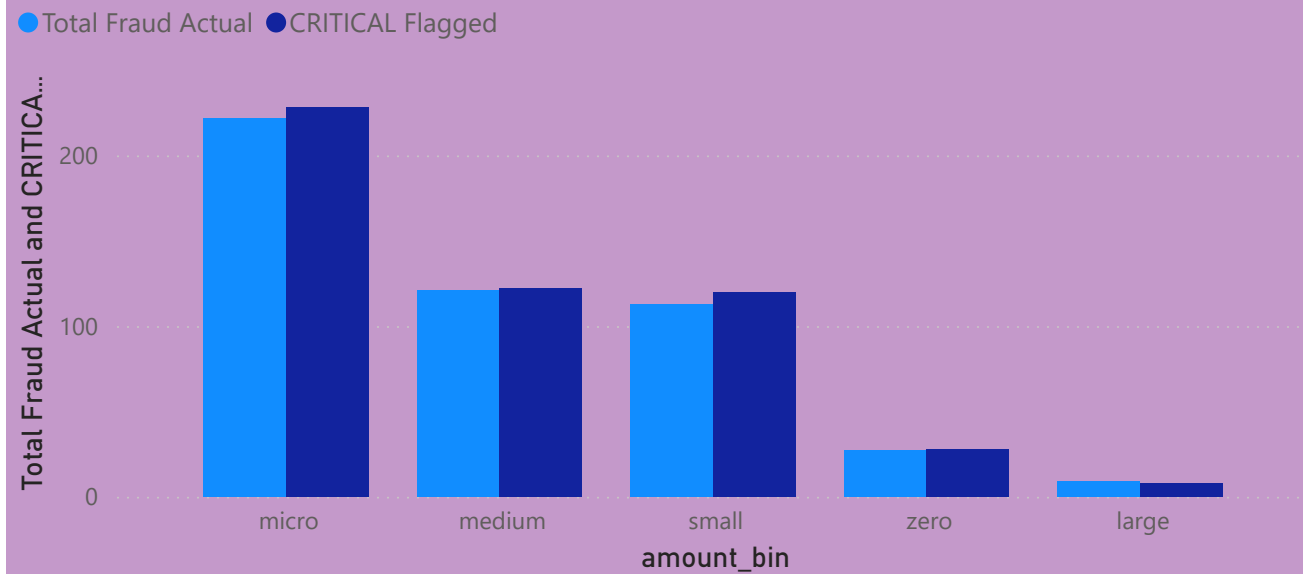
94.47

CRITICAL Precision %

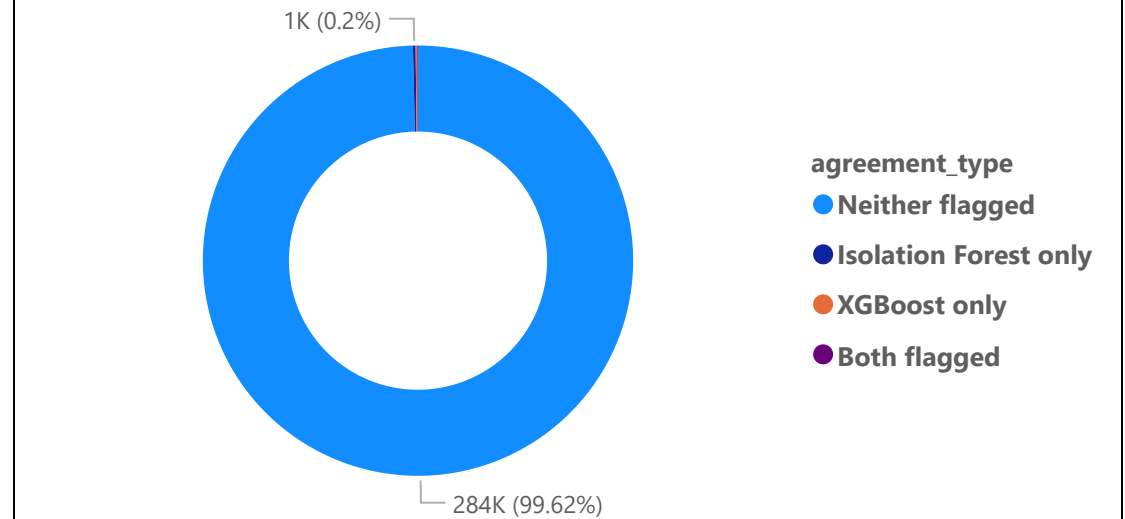
51.24K

Amount at Risk

Total Fraud Actual and CRITICAL Flagged by amount_bin



Sum of transaction_count by agreement_type



- **TP** (True Positives / Fraud Caught) = **84**
- **FP** (False Positives / False Alarms) = **14**
- **FN** (False Negatives / Missed Fraud) = **14**
- **TN** (True Negatives / Correct Legit) = **56,833**

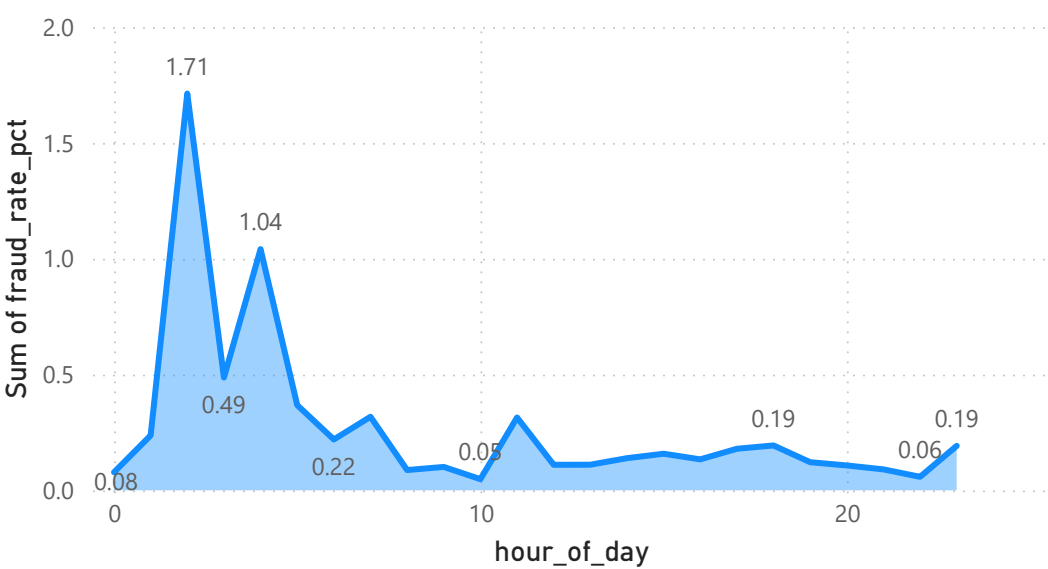
hour_risk_band

☐ HIGH_RISK_HOURS

☐ STANDARD_HOURS

2	0.31	0.14
Peak Fraud Hour	Night Hour Fraud Rat...	Day Hour Fraud Rate %

Sum of fraud_rate_pct by hour_of_day



hour_of_day	time_of_day_label	Sum of total_txns	Sum of actual_fraud_count	Sum of fraud_rate_pct	hour_risk_band
0	Late night (0-5am)	23934	124	3.92	HIGH_RISK_HOURS
1	Late night (0-5am)	23934	124	3.92	HIGH_RISK_HOURS
2	Late night (0-5am)	23934	124	3.92	HIGH_RISK_HOURS
3	Late night (0-5am)	23934	124	3.92	HIGH_RISK_HOURS
4	Late night (0-5am)	23934	124	3.92	HIGH_RISK_HOURS
5	Late night (0-5am)	23934	124	3.92	HIGH_RISK_HOURS
6	Morning (6-11am)	70912	118	1.09	STANDARD_HOURS
7	Morning (6-11am)	70912	118	1.09	STANDARD_HOURS
8	Morning (6-11am)	70912	118	1.09	STANDARD_HOURS
9	Morning (6-11am)	70912	118	1.09	STANDARD_HOURS
10	Morning (6-11am)	70912	118	1.09	STANDARD_HOURS
11	Morning (6-11am)	70912	118	1.09	STANDARD_HOURS
12	Afternoon (12-5pm)	96435	134	0.83	STANDARD_HOURS
Total		284807	492	6.61	

94.47

CRITICAL Precision %

5.53

False Positive Rate %

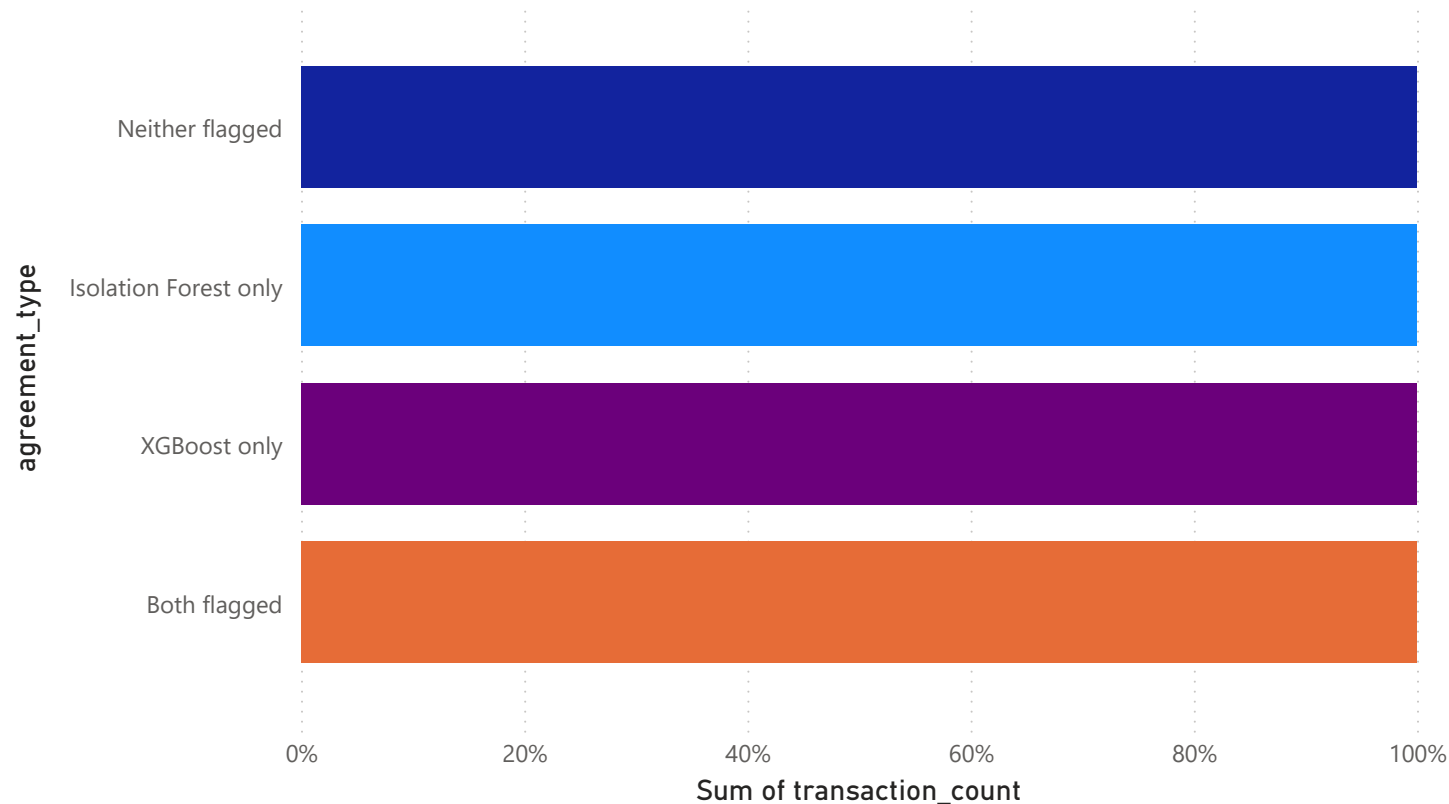
0.00

ISO Only Catch %

0% on historical data —
expected. Isolation Forest is
retained to monitor the live
stream for novel fraud patterns
that the supervised model
hasn't seen yet.

Sum of transaction_count and Sum of actual_fraud by agreement_type and actual_fraud

actual_fraud ● 0 ● 14 ● 133 ● 345



On pre-processed data where fraud is already
geometrically separated, XGBoost is sufficient
alone. Isolation Forest is retained in the streaming
layer as an early-warning system for novel fraud
patterns — the kind that appear in a live card
stream before we have labelled examples to train
XGBoost on.

4K

Txns Last 5 Min

1

CRITICAL Last 5 Min

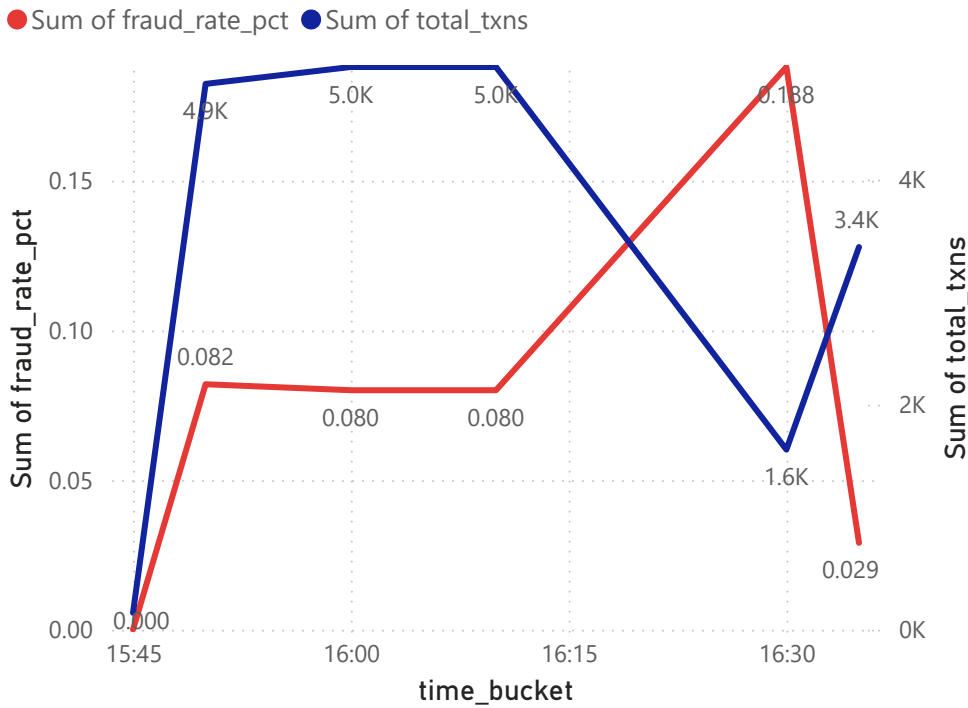
2

Block Now 30 Min

0.10

Avg Score Last 5 Min

Live Fraud Rate — Rolling 1 Hour



alert_time	Sum of Amount	Sum of fraud_score_pct	action_required	Sum of hour_of_day
13-05-2026 16:34:41	529.00	99.50	BLOCK_NOW	0
Total	529.00	99.50		0